PYTHON ACTIVITY-BASED LEARNING (ABL)

CASE STUDY 1: PROGRESSIVE ELECTRICITY BILLING

1. Problem Statement

Write a Python program to calculate an electricity bill based on the number of units consumed by a household. Different rates are applied to different unit ranges.

2. Objectives

1. To calculate an electricity bill using Python.
2. To understand if, elif, and else statements.
3. To use variables and arithmetic operators.
4. To display the final electricity bill.

3. Assumptions

• First 100 units: ■2 per unit.

• Next 100 units: ■3 per unit.

• Above 200 units: ■5 per unit.

• No additional charges or taxes are included.

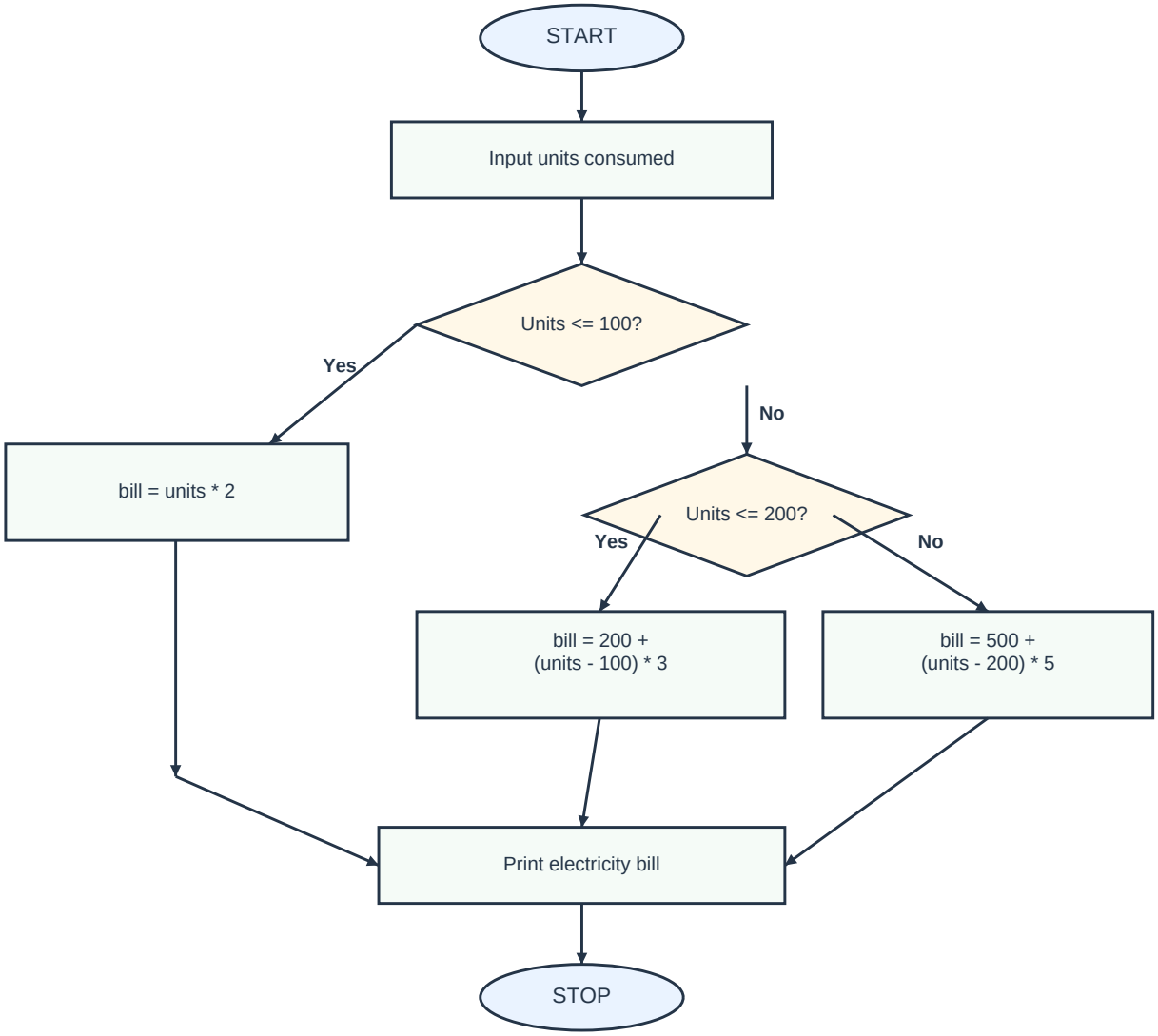
Note: These are assumed rates for this assignment, not official electricity tariffs.

4. Algorithm

1. Start the program.
2. Enter the number of electricity units consumed.
3. If units are less than or equal to 100, bill = units × 2.
4. If units are between 101 and 200, bill = 200 + (units – 100) × 3.
5. If units are above 200, bill = 500 + (units – 200) × 5.
6. Display the electricity bill.
7. Stop the program.

5. Flowchart (Drawing Method)

This flowchart uses standard shapes and shows the Yes/No decision branches.



Oval: Start/Stop | Parallelogram-style step: Input/Output | Diamond: Decision | Box: Calculation

6. Pseudocode

```
START
INPUT units

IF units <= 100 THEN
    bill = units * 2
ELSE IF units <= 200 THEN
    bill = 200 + (units - 100) * 3
ELSE
    bill = 500 + (units - 200) * 5
END IF

PRINT bill
STOP
```

7. Python Source Code

```
units = int(input("Enter electricity units: "))

if units <= 100:
    bill = units * 2
elif units <= 200:
    bill = 200 + (units - 100) * 3
else:
    bill = 500 + (units - 200) * 5

print("Electricity Bill =", bill)
```

8. Test Cases

Test Case	Units Consumed	Expected Bill
1	50	■100
2	100	■200
3	150	■350
4	200	■500
5	250	■750

9. Sample Output

```
Enter electricity units: 150
Electricity Bill = 350

Enter electricity units: 250
Electricity Bill = 750
```

10. Conclusion

Through this case study, I learned how to calculate an electricity bill using Python. I understood how to use variables, arithmetic operators, and conditional statements such as if, elif, and else.